

MATHS — SEMESTER - I

8 lesson(s) included, 1 missing

Included:

- Ch-1 :Numbers
- Ch-3 :Negative Numbers and Integers
- Ch-5 :Fractions and Decimals
- Ch-6 :Playing with Numbers
- Ch-9 :Speed,Distance and Time
- Ch-10 :Algebra(K.C-1,2,3)
- Ch-11 :Basic Geometrical Ideas 2-D
- Ch-16 :Data Handling (K.C1,2), Graph-Plotting points,Ch14:Construction s (K.C-1,2)

MISSING (no downloaded Question Bank found):

- Ch-8 :Percentage

Question:1

Multiple choice question:

- (i) The successor of the greatest 5 – digit numbers is _____
(a) 100000 (b) 999999 (c) 10000 (d) 99999
- (ii) The face value of the digit 8 in the number 2,56,87,630 is _____
(a) 8 (b) 800 (c) 80,000 (d) 87,000
- (iii) The estimated value of $36+71-55$ is _____
(a) 40 (b) 50 (c) 70 (d) 150
- (iv) The numeral 27348 rounded off to the nearest thousand is _____
(a) 27000 (b) 30000 (c) 25000 (d) 20000
- (v) Assertion(A): The place value of 7 in the numeral 976058 is 7000
Reason(R): The place value of a digit in a number depends upon the place it occupies in the place value chart
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is the not correct explanation of A.
(c) A is true , but R is false.
(d) A is false ,but R is true.

Question :2

Solve the following:

- (i) Find the difference between the greatest and the least numbers that can be writtern using the digits 5,2,4,6,9 each only once .
- (ii) Find the product of the place value and face value of 6 in 87506423.
- (iii) Sandhya had 187 marbles. She gave 28 marbles to her brother. How many marbles are left with sandhaya? Estimate your answer to the nearest ten.
- (iv) Write in short form:
a) $4 \times 1,00,000 + 5 \times 10,000 + 3 \times 1000 + 2 \times 100 + 5 \times 10 + 1$
b) $5 \times 10000 + 4 \times 1000 + 7$
- (v) Write the number 36578902 in both the Indian and International system of numeration using commas. Also write their number names.
- (vi) There are 422 eggs for packing in boxes . If 12 eggs are packed in one box , estimate how many boxes are needed to pack all eggs?

(vii) Estimate the following by rounding off each number to its greatest number

(i) 161×68

(ii) $7564 - 4341$

(iii) $97 \div 18$

(iv) $6708 + 543$

(viii) A pair of shoes costs Rs.791. What will be the cost of 9 such pairs of shoes?
Give your answer by estimating .

(ix) A bag contains rice weighs 25 kg 300 g .Express the weight :

(a) In grams

(b) in milligrams

(x) Niru bought a new hot water bottle that holds 1,500 millilitres of water. How many litres is that?

Ch-3:Negative Number and Integer

- 1) Write the integers that lie between (i) -5 and 4 , (ii) -2 and 10
- 2) Which is greater (i) -7 or -3 (ii) 5 and -3
- 3) Arrange the following integers in ascending order:
(i) $5, -2, 0, -8, 3, -1$ (ii) $-12, -5, 4, -1, 9, 0$
- 4) Add:
(i) $(-8) + 5$ (ii) $7 + (-12)$
(iii) $(-6) + (-9)$ (iv) $15 + 23$
(v) $(-10) + (-6)$ (vi) $(-7) + 8$
- 5) Represent -4 , 0 and $+5$ on a number line.
- 6) The temperature in a town was -3°C in the morning. By afternoon, it increased by 8°C . What was the afternoon temperature?
- 7) A lift is on the 3rd floor and goes down 7 floors. On which floor does it stop?
- 8) A submarine is 40 m below sea level. It rises 15 m. What is its new position relative to sea level?
- 9) Simplify: (i) $(-15) + 9 + 4$ (ii) $(-7) + (-8) - (-5)$
(iii) $13 + (-12) - 5$ (iv) $5 - (-7) + 4$
- 10) Use a number line to find:
(i) $(-3) + 5$ (ii) $4 + (-7)$ (iii) $(-6) + (-4)$
- 11) Subtract : (i) 8 from 12 (ii) 18 from (-12)
(iii) (-15) from 14 (iv) (-11) from (-7)
- 12) Write the Integers: (i) 12 m below sea level
(ii) A profit of Rs.500
(iii) 6 steps down

1) Identify the type of fraction:

a) $\frac{7}{4}$ b) $\frac{3}{8}$ c) $2\frac{3}{5}$

2) Convert into mixed numbers:

a) $\frac{17}{5}$ b) $\frac{29}{6}$ c) $\frac{45}{8}$

3) Convert into improper fractions:

a) $4\frac{1}{2}$ b) $6\frac{2}{3}$ c) $9\frac{2}{5}$

4) Reduce to lowest terms:

a) $\frac{24}{36}$ b) $\frac{45}{60}$ c) $\frac{18}{48}$

5) Find equivalent fractions (any two) for:

a) $\frac{2}{3}$ b) $\frac{5}{7}$ c) $\frac{3}{5}$

6) Compare using <, >, or =:

a) $\frac{3}{5}$ — $\frac{4}{7}$ b) $\frac{7}{12}$ — $\frac{5}{8}$ c) $\frac{2}{3}$ — $\frac{8}{12}$

7) Arrange in ascending order:

a) $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ b) $\frac{3}{5}, \frac{1}{4}, \frac{7}{10}$ c) $\frac{5}{6}, \frac{2}{3}, \frac{3}{4}$

8) Add the following fractions:

a) $\frac{2}{5} + \frac{3}{10}$ b) $1\frac{3}{4} + 2\frac{1}{2}$ c) $\frac{3}{8} + \frac{5}{6}$

9) Subtract the following fractions:

a) $\frac{7}{8} - \frac{3}{4}$ b) $5\frac{1}{3} - 2\frac{1}{4}$ c) $\frac{9}{10} - \frac{2}{5}$

10) Multiply the following fractions:

a) $\frac{2}{3} \times \frac{4}{5}$

b) $1\frac{2}{3} \times 3\frac{1}{2}$

c) $\frac{5}{6} \times 12$

11) Divide the following fractions:

a) $\frac{3}{4} \div \frac{2}{3}$

b) $4\frac{1}{5} \div 1\frac{2}{5}$

c) $\frac{7}{8} \div 14$

12) A ribbon is $3\frac{3}{5}$ m long. If $1\frac{4}{5}$ m is cut off, what length remains?

13) Rita ate $\frac{2}{5}$ of a cake and Meena ate $\frac{1}{4}$ of it. What fraction did they eat together?

14) A tank is $\frac{3}{4}$ full. If it holds 60 litres when full, how many litres are in it now?

15) Write in decimal form:

a) $\frac{7}{10}$

b) $\frac{23}{100}$

c) $5\frac{3}{4}$

16) Write in fraction form (lowest terms):

a) 0.6

b) 0.75

c) 2.05

17) Arrange in ascending order:

a) 3.4, 3.04, 3.44

b) 0.5, 0.55, 0.505

c) 2.1, 2.01, 2.11

18) Add the following decimals:

a) $12.5 + 3.75$

b) $0.6 + 2.45 + 1.008$

c) $7.2 + 15.36$

19) Subtract the following decimals:

a) $15.6 - 8.75$

b) $20 - 4.65$

c) $9.03 - 2.147$

20) Simplify :

a) $2.87 + 3.456 - 4.5$

b) $0.004 + 2.334 - 1.1$

c) $1.003 - 0.5 + 3.6$

Class:VI

I. Simplify:

1. $19 - (1 + 5) - 3$
2. $30 \times 6 \div (5 - 2)$
3. $28 - (3 \times 8) \div 6$
4. $9 - [(4 - 3) + 2 \times 5]$
5. $[18 - (15 \div 5) + 6]$
6. $[(4 \times 2) - (4 \div 2)] + 8$
7. $60 - \{16 \div (4 \times 6 - 8)\}$
8. $15 - [16 - \{12 + 21 \div (9 - 2)\}]$

II. Write all the factors of:

- (i) 16 (ii) 21 (iii) 39 (iv) 48 (v) 64 (vi) 98

III. Write the first six multiples of:

- (i) 4 (ii) 9 (iii) 11 (iv) 15 (v) 18 (vi) 16

IV. Solve:

- a) The product of two numbers is 36 and their sum is 13. Find the numbers.
- b) The product of two numbers is 48 and their sum is 16. Find the numbers.
- c) Write two numbers which differ by 3 and whose product is 54.
- d) Without making any actual division show that 7007 is divisible by 7.
- e) Without making any actual division show that 2300023 is divisible by 23
- f) Without making any actual division, show that each of the following numbers is divisible by 11
 - (i) 11011
 - (ii) 110011
 - (iii) 11000011

g) Without actual division, show that each of the following numbers is divisible by 8

(i) 1608 (ii) 56008 (iii) 240008

h) Find which of the following numbers are divisible by 2:

(i) 352 (ii) 523 (iii) 496 (iv) 649

i) Find which of the following number are divisible by 4:

(i) 222 (ii) 532 (iii) 678 (iv) 9232

j) Find which of the following numbers are divisible by 8:

(i) 324 (ii) 2536 (iii) 92760 (iv) 444320

k) Find which of the following numbers are divisible by 3:

(i) 221 (ii) 543 (iii) 28492 (iv) 92349

l) Find which of the following numbers are divisible by 5:

(i) 5080 (ii) 66666 (iii) 755 (iv) 9207

m) Find which of the following numbers are divisible by 11:

(i) 5918 (ii) 68,717 (iii) 3882 (iv) 10857

n) In each of the following numbers, replace M by the smallest number to make resulting number divisible by 3:

(i) 64 M 3 (ii) 46 M 46 (iii) 27 M 53

V) Find:

1) The numbers 13 and 31 are prime numbers. Both these numbers have the same digits, 1 and 3. Find such pairs of prime numbers up to 100.

2) Express the following as the sum of two odd primes.

(a) 44 (b) 36 (c) 24 (d) 18

3) Which of the following numbers are prime?

(a) 23 (b) 51 (c) 37 (d) 26

4) Write five pairs of prime numbers less than 20 whose sum is divisible by 5. (Hint: $3 + 7 = 10$)

5) Which of the following numbers are co-prime?

(a) 18 and 35 (b) 15 and 37 (c) 30 and 415 (d) 17 and 68 (e) 216 and 215

VI Find the HCF of the following numbers using prime factorisation method:

(a) 186, 480

(b) 300, 420

(c) 1928, 602

(d) 279, 639

(e) 362, 84

VII a) Determine the greatest 3-digit number exactly divisible by 80, 10 and 120

VIII Find the HCF using long division method.

a) 272,950 b) 754, 144, 180 c) 504,480

IX) Solve:

a) Three tankers contain 403 litres, 434 litres and 465 litres of diesel, respectively. Find the maximum capacity of a container that can measure the diesel of the three containers the exact number of times.

b) The length, breadth and height of a room are 825 cm, 675 cm and 450 cm, respectively. Find the longest tape that can measure the room's three dimensions exactly.

X) Which of the following numbers are co-prime?

(a) 18 and 35 (b) 15 and 37 (c) 30 and 415 (d) 17 and 68 (e) 216 and 215

(f) 81 and 16

XI) a) Find the HCF of 182,227,347 leaves remainder 2 in each case

b) Find the HCF of 886 and 3158 leaves remainder 4 and 8 respectively

XII) Reduce the following fractions to the lowest terms .

(a) $\frac{48}{60}$ (b) $\frac{150}{60}$ (c) $\frac{84}{98}$ (d) $\frac{12}{52}$ (e) $\frac{7}{28}$

XIII) Find all the prime factors of 1729, 4641 and arrange them in exponential form.

XIV)Find:

- a) Determine the greatest 3-digit number exactly divisible by 8, 10 and 12.
- b) Find the least number, which, when divided by 6, 15 and 18 leave the remainder 5 in each case.
- c) Find the smallest 4-digit number, which is divisible by 18, 24 and 32.

XV)Find the LCM by the division method.

- a) 8,16,24,36 ii) 32,36,40 and 45

XVI) Find the LCM by prime factorisation method.

- a) 21, 49,63, 56 b) 45,60,15 and 90

XVII)Solve:

- a) Three boys step off together from the same spot. Their steps measure 63 cm, 70 cm and 77 cm, respectively. What is the minimum distance each should cover so that all can cover the distance in complete steps?
- b) The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds, respectively. If they change simultaneously at 7 a.m., at what time will they change simultaneously again?

XVIII) a)Prove that $\text{LCM}(12, 7) \times \text{HCF}(12, 7) = \text{Product}(12, 7)$.

b)HCF and LCM of two numbers are 13 and 1989 respectively. If one number is 221, determine the other.

c) Find the product of HCF and LCM of two numbers 50 and 95

XIX) Mental Maths :

- 1) Which of the following are not twin-primes?
(a) 3, 5 (b) 5, 7 (c) 11, 13 (d) 17, 23
- 2) The total number of even prime numbers is
(a) 0 (b) 1 (c) 2 (d) unlimited

- 3) Which of the following numbers is divisible by 9?
(a) 9076185 (b) 92106345 (c) 10349576 (d) 95103476
- 4) If $1*548$ is divisible by 3, then * can take the value
(a) 0 (b) 2 (c) 7 (d) 8
- 5) The sum of the prime numbers between 60 and 75 is
(a) 199 (b) 201 (c) 211 (d) 272
- 6) The HCF of two consecutive natural numbers is
(a) 0 (b) 1 (c) 2 (d) non-existent
- 7) The LCM of 24, 36 and 40 is
(a) 4 (b) 90 (c) 360 (d) 720
- 8) If the HCF of two number is 16 and their product is 3072, then their LCM is
(a) 182 (b) 192 (c) 12 (d) None of these
- 9) Which of the following number is a multiple of 6?
(a) 24 (b) 32 (c) 45 (d) 53
- 10) What are the numbers which are multiples of 2 called?
(a) Odd numbers (b) Even numbers (c) Prime numbers (d) Composite numbers

XX) State whether the following statements are True or False:

- (a) The sum of three odd numbers is even.
- (b) The sum of two odd numbers and one even number is even.
- (c) The product of three odd numbers is odd.
- (d) If an even number is divided by 2, the quotient is always odd.
- (e) All prime numbers are odd.
- (f) Prime numbers do not have any factors.
- (g) Sum of two prime numbers is always even.
- (h) 2 is the only even prime number.
- (i) All even numbers are composite numbers.
- (j) The product of two even numbers is always even.

The Vikasa School, Tuticorin.
Question bank
Subject: Mathematics
Ch – 9 Speed, Distance and Time

Class: VI

1.A train travels 700 metres in 35 seconds. What is its speed in km/h?

2.Rohit goes 350 km in 5 hours. Find:

(i) his speed

(ii) the distance covered by Rohit in 6.2 hours

(iii) the time taken by him to cover 210 km

3. A boy swims 200 m in still water and then returns back to the point of start in total 10 minutes. Find the speed of his swim in

(i) ms^{-1}

(ii) kmh^{-1}

4.A train covers 51 km in 3 hours. Calculate its speed. How far does the train go in 30 minutes?

5. Express the following in m/min.

i. 60 km/hr

ii. 95 km/hr

iii. 36 km/hr

6. Express the following in m/sec.

i. 43 m/hr

ii. 24 m/min

iii. 82 m/hr

iv. 6 km/hr

v. 40 km/hr

vi. 90 km/hr

vii. 120 km /hr

viii.300 km/hr

6. Complete the following table.

Q.No	Distance	Time	Speed
1	360 km	-	90 km/hr
2	840 km	7 hrs	
3		12 hrs	60km /hr
4	7200 m	360 secs	

7. Determine the time taken when, distance is 7150 km and speed is 780 km/hr.

8.If distance travelled by a train is 495 km in 4 hours 30 minutes, what is its speed?

9. A cyclist covers 950 m in 5 minutes. Find his speed in km/ hour.

10.The speed of the train is 72 km per hour. Find its speed in metre per second.

11. Express the speed of 60 m per minute in km per hour.

12. A bike covers a distance of 705 km in 15 hours. Find the speed of the bike.

13. A truck covers 450 km in 9 hours and a bike covers 525 km in 10 hours. Which moves faster? Find the speed of both.

14. A man runs at the speed of 10 km/hr. How much time will he take to cover 750 metres?

15. Aaron ran 500 metre in 100 Seconds. Find the speed in km per hour.

16. Find the speed in the following cases.

Distance Covered

Time Taken

i. 285 km

5 hrs

ii. 365 km

3 hrs

iii. 504 km

7 hrs

iv. 1245 km

15 hrs

17. MCQS

i) Which of the following is true when we convert m s^{-1} into km h^{-1} ?

- (a) Multiply by $\frac{18}{5}$ (b) Multiply by $\frac{5}{18}$ (c) Multiply by 1 (d) Multiply by 3

ii) Teacher asked in the class that is what is the value of 35 m s^{-1} into km h^{-1} ?

- (a) 154 km h^{-1} (b) 148 km h^{-1} (c) 126 km h^{-1} (d) 130 km h^{-1}

iii) Question asked in the examination that if Amit runs 400 metre in 50 seconds. Find his speed?

- (a) 4 m s^{-1} (b) 2 m s^{-1} (c) 8 m s^{-1} (d) 5 m s^{-1}

iv) A 120 m long train is travelling at a speed of 36 km h^{-1} , find the time taken by the train to pass a telegraph post?

- (a) 14 sec (b) 13 sec (c) 12 sec (d) 18 sec

v) A train whose length is 200 m, passes a telegraph pole in 10 sec. Find the speed of the train in km h^{-1} ?

- (a) 72 km h^{-1} (b) 74 km h^{-1} (c) 75 km h^{-1} (d) 76 km h^{-1}

vi) Which of the following relation between speed, distance and time is true?

(a) $\text{Speed} = \text{Distance} \times \text{time}$

(b) $\text{Speed} = \frac{\text{Time}}{\text{Distance}}$

(c) $\text{Time} = \frac{\text{Distance}}{\text{Speed}}$

(d) $\text{Distance} = \frac{\text{Speed}}{\text{Time}}$

vii) For the question below, choose the correct option from the following standard rules:

(a) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is NOT the correct explanation of Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

Assertion (A): A train covering a distance of 150 km in 3 hours runs at a speed of 50 km/h.

Reason (R): $\text{Speed} = \frac{\text{Time}}{\text{Distance}}$

Question – 1 Multiple Choice Questions :

1. Which of the following is an algebraic expression?

- a) $5 + 3 = 8$ b) $7x + 2$ c) $12 > 5$ d) 8

2. The mathematical form of “Ten divided by x equals four” is:

- a) $10x = 4$ b) $x \div 10 = 4$ c) $10/x = 4$ d) $10 + x = 4$

3. The exponent form of $a \times a \times a \times a \times a$ is:

- a) $5a$ b) a^5 c) $a + 5$ d) a^2

4. If one square requires 4 matchsticks and each additional square shares one side, then 3 squares require:

- a) 10 b) 12 c) 9 d) 15

5. Assertion and Reason

Assertion (A): $5x + 3y - 4z$ is an algebraic expression.

Reason (R): An algebraic expression is formed by combining constants and variables using mathematical operations.

- a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

Question – 2 :

1. Write the following using mathematical symbols:

- a) Seven increased by two
b) Nine taken from x

2. Write in words:

- a) $x + y$ b) $5x$

3. Convert into algebraic form:

a) Three times a number x

b) Six less than y

4. Write in exponent form:

a) $p \times p \times p \times p$

b) $x \times x \times x \times y \times y$

5. Write the coefficient of:

a) x in $7x$

b) y in $12y$

6. State whether the following are monomials:

a) $8x^2y$

b) $5x + 3$

7. One box contains 12 candies. Write an expression for the number of candies in n boxes.

8. A mother gives 4 toffees to each child. Write an expression for the number of toffees needed for x children.

9. Five students stand in each row. Write an expression for the number of students in y rows.

10. Rama's age is x years. Her father is 4 years more than 5 times her age. Write an expression for her father's age.

11. Rewrite using symbols:

a) x exceeds y by 9

b) The product of p and q divided by s

c) Four times m is greater than six

12. Write in words:

a) $3(x + 2)$

b) $(x + z) / y$

c) $x < y < z$

13. Matchstick Pattern – Letter V

a) Find the number of matchsticks required to make 4 letter V patterns.

b) Find the number of matchsticks required to make 10 letter V patterns.

c) Write a general rule in terms of n .

14. Matchstick Pattern – Squares

A row of squares is formed using matchsticks.

Number of Squares : 1, 2, 3

Matchsticks : 4, 7, 10

- a) Find the number of matchsticks required for 4 squares.
- b) Find the number of matchsticks required for 6 squares.
- c) Write a general rule for n squares.

15. Matchstick Pattern – Hexagons

- a) Find the number of matchsticks required for 1, 2 and 3 hexagons placed in a row.
- b) Write the algebraic rule for n hexagons.

16. Kinjal is 4 years older than Reema.

- a) If Reema's age is x years, express Kinjal's age in terms of x .
- b) Find Kinjal's age when Reema is 11 years old.

17. Mamta's maths marks are 35 more than Sunita's marks.

- a) If Sunita's marks are m , write an expression for Mamta's marks.
- b) Find Mamta's marks when Sunita scores 48 marks.

18. A question paper contains x objective questions and y subjective questions.

Each objective question carries 2 marks and each subjective question carries 4 marks.

- a) Write an expression for the total marks.
- b) Find the total marks when $x = 10$ and $y = 5$.

19. A cuboidal box has height h cm which is 60 cm less than its length l cm.

- a) Write the relation between h and l .
- b) Find the height when $l = 150$ cm.

20. Three boys are climbing steps.

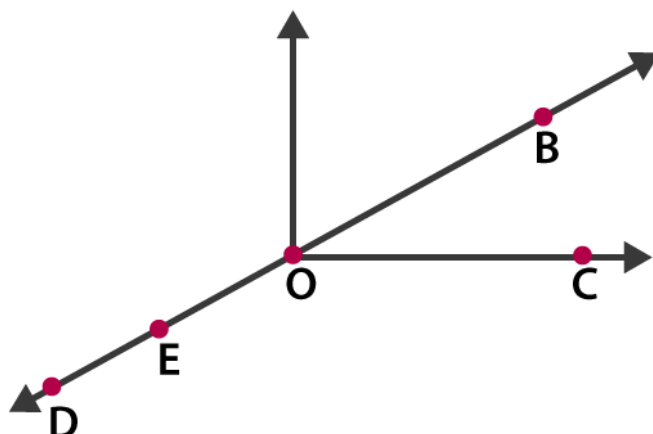
The first boy is at step x . The second boy is 5 steps ahead and the third boy is 8 steps behind.

- a) Write expressions for the positions of the second and third boys.
- b) Find their positions when $x = 20$.

Class:VI

1. Use the figure to name:

- (a) Five points
- (b) A line
- (c) Four rays
- (d) Five line segments

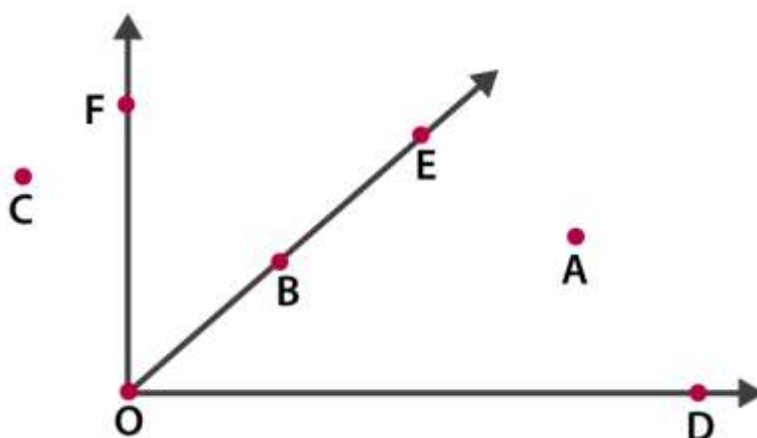


2. Name the line given in all possible (twelve) ways, choosing only two letters at a time from the four given.



3. In the given diagram, name the point(s)

- (a) In the interior of $\angle DOE$
- (b) In the exterior of $\angle EOF$
- (c) On $\angle EOF$



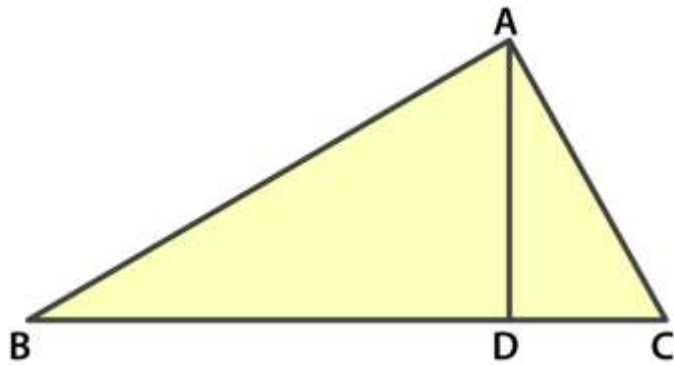
4. Draw a rough sketch of a triangle ABC. Mark a point P in its interior and a point Q in its exterior. Is the point A in its exterior or in its interior?

5. (a) Identify three triangles in the figure.

(b) Write the names of seven angles.

(c) Write the names of six line segments

(d) Which two triangles have $\angle B$ as common?



6. From the figure, identify and name them.

(a) the centre of circle

(b) three radii

(c) a diameter

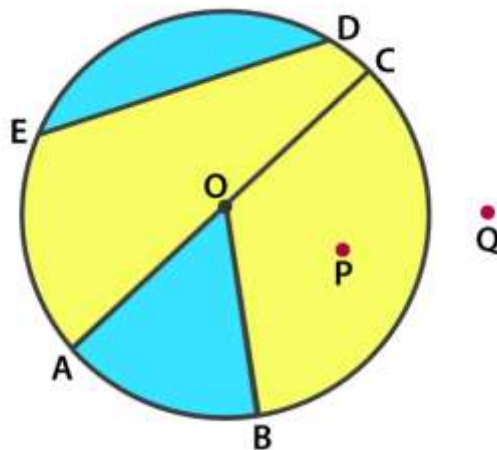
(d) a chord

(e) two points in the interior

(f) a point in the exterior

(g) a sector

(h) a segment



7. Draw any circle and mark

(a) its centre

(b) a radius

(c) a diameter

(d) a sector

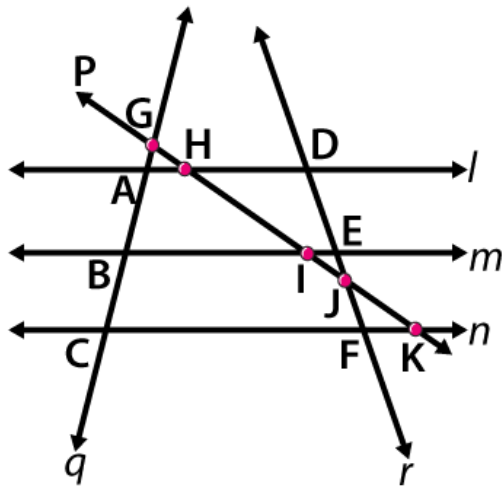
(e) a segment

(f) a point in its interior

(g) a point in its exterior

(h) an arc

7. From the below figure, write



(i) all pairs of parallel lines.

(ii) all pairs of intersecting lines.

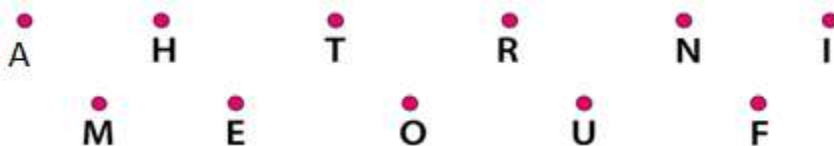
(iii) lines whose point of intersection is I.

(iv) lines whose point of intersection is D.

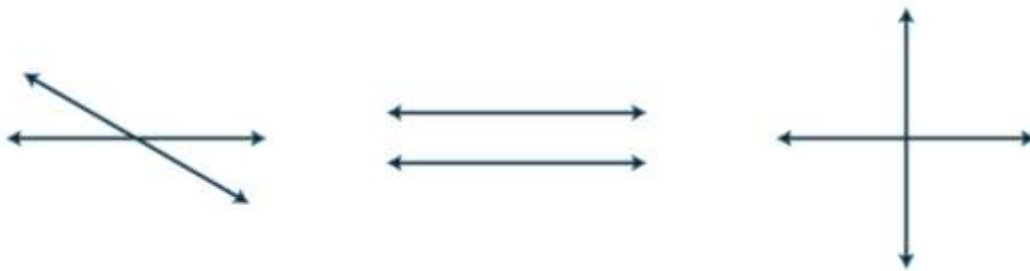
(v) lines whose point of intersection is E.

(vi) lines whose point of intersection is A.

8. In given figure, points are given in two rows. Join the points AM, HE, TO, RUN, IF. How many line segments are formed?



9. Classify the following lines as parallel or intersecting?



10. Multiple choice questions

1. Which of the following has a definite length?

A) Line B) Ray C) Line segment D) None of these

2. How many lines can pass through a single given point?

A) 1 B) 2 C) 3 D) Infinite

3. Which of the following 2D figures is a simple closed curve made up of line segments?

A) Circle B) Polygon C) Angle D) Ray

4. A quadrilateral has how many diagonals?

A) 1 B) 2 C) 3 D) 4

5. What is the perimeter (boundary length) of a circle called?

A) Diameter B) Radius C) Circumference D) Chord

6. A diameter of a circle passes through the center and is equal to:

A) $2 \times \text{Radius}$ B) $\text{Radius} \div 2$ C) $3 \times \text{Radius}$ D) Radius itself

7. **Assertion (A):** A triangle is a polygon made of three-line segments.

Reason (R): A polygon is a simple closed curve made entirely of line segments.

a) Both A and R are true and R is the correct explanation of A

b) Both A and R are true but R is not the correct explanation of A

c) A is true but R is false

d) A is false but R is true

8. **Assertion (A):** A line extends indefinitely in both directions and has an infinite number of points.

Reason (R): A line segment is a part of a line with two definite endpoints.

a) Both A and R are true and R is the correct explanation of A

b) Both A and R are true but R is not the correct explanation of A

c) A is true but R is false

d) A is false but R is true

1. In a study of number of accidents per day, the observations for 30 days were obtained as follows:

6 3 5 6 4 3 2 5 4 2

4 0 5 3 6 1 5 5 2 6

2 1 2 2 0 5 4 6 1 6

Construct a suitable frequency distribution table.

2. The following data represents the weekly wages (in Rs) of 15 workers in a factory:

900, 850, 800, 850, 800, 750, 950, 900, 950, 800, 750, 900, 750, 800 and 850

Prepare a frequency distribution table. Now find,

(i) how many workers are getting less than Rs 850 per week?

(ii) how many workers are getting more than Rs 800 per week?

3. Using the data, given below, construct a frequency distribution table:

9, 17, 12, 20, 9, 18, 25, 17, 19, 9, 12, 9, 12, 18, 17, 19, 20, 25, 9 and 12. Now answer the following:

(i) How many numbers are less than 19?

(ii) How many numbers are more than 20?

(iii) Which of the numbers, given above, is occurring most frequently?

4. Marks scored by 30 students of class VI are as given below:

38, 46, 33, 45, 63, 53, 40, 85, 52, 75, 60, 73, 62, 22, 69, 43, 45, 33, 47, 41, 29, 43, 37, 49, 83, 44, 55, 22, 35 and 45. State:

(i) the highest marks scored

(ii) the lowest marks scored

(iii) the range of marks

5. The number of animals in five villages are as follows:

Village	A	B	C	D	E
No. of animals	160	240	180	80	120

Prepare a pictograph of these animals using one symbol to represent 20 animals.

6. The following pictograph shows different subject books which are kept in a school library.

Subject	Number of books
Hindi	       
English	        
Math	      
Science	     
History	   

Taking symbol of one book = 50 books, find:

- (i) how many History books are there in the library?
- (ii) how many Science books are there in the library?
- (iii) Which books are maximum in number?
- (iv) How many more English books are there in library than History?
- (v) Find the total number of books in the library.

7. The sale of vehicles, in a particular city, during the first six months of the year 2016 is shown below:

Key :



Month	Number of vehicles sold
Jan	     
Feb	    
March	       
April	 
May	  
June	      

Using the pictograph, drawn above, answer the following questions:

- (i) How many more Vehicles has sold on March as compared to May?
- (ii) What is the total number of vehicles sold by all the dealers?